BCS quantum simulation: paper reproduction

Notebook reprodutível para acompanhar o manuscrito `papers/bcs_quantum_simulation.md`.

Ele assume que os experimentos já foram executados e que as tabelas/figuras existem em `results/`. A última seção traz os comandos para regenerar tudo.

In [1]:
```
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import Image, display, Markdown

cwd = Path.cwd().resolve()
ROOT = cwd if (cwd / 'results').exists() else cwd.parent
TABLES = ROOT / 'results' / 'tables'
FIGURES = ROOT / 'results' / 'figures'

pd.set_option('display.max_columns', 80)
pd.set_option('display.precision', 6)
```

Parâmetros principais

O caso padrão usado no paper é o Hamiltoniano BCS reduzido com quatro modos de pares.

In [2]:
```
params = {
    'xi': [-1.0, -0.35, 0.35, 1.0],
    'g': 0.8,
    'occupied': [0, 1],
    'time': 2.5,
    'final_time_observables': 6.0,
    'trotter_steps': 8,
    'trotter_order': 2,
}
params
```

Out[2]:
```
{'xi': [-1.0, -0.35, 0.35, 1.0],
 'g': 0.8,
 'occupied': [0, 1],
 'time': 2.5,
 'final_time_observables': 6.0,
 'trotter_steps': 8,
 'trotter_order': 2}
```

Funções auxiliares

In [3]:
```
def read_table(name):
    path = TABLES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing table: {path}')
    return pd.read_csv(path)

def show_figure(name, width=850):
    path = FIGURES / name
    if not path.exists():
        raise FileNotFoundError(f'Missing figure: {path}')
    display(Image(filename=str(path), width=width))

def section(title):
    display(Markdown(f'## {title}'))
```

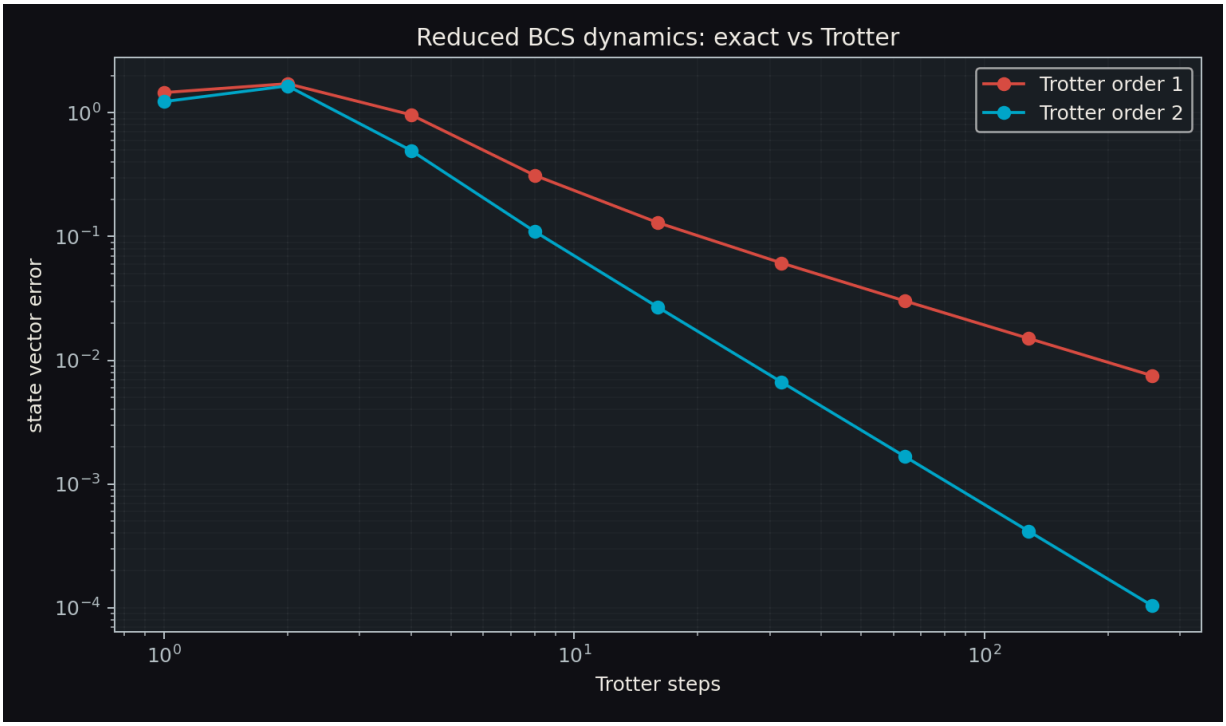
Dinâmica BCS e erro de Trotter

In [4]:
```
trotter = read_table('bcs_dynamics_trotter.csv')
display(trotter)
best = trotter.sort_values('fidelity', ascending=False).iloc[0]
best[['time', 'steps', 'order', 'fidelity', 'state_error', 'exact_gap_proxy'
```

	time	steps	order	fidelity	state_error	exact_energy	trotter_energy	exact_pair_r
0	2.5	1.0	1.0	0.016402	1.459181	2.7	-2.723483	
1	2.5	2.0	1.0	0.422325	1.720592	2.7	0.591860	
2	2.5	4.0	1.0	0.331926	0.964730	2.7	1.138241	
3	2.5	8.0	1.0	0.905103	0.313054	2.7	2.725392	
4	2.5	16.0	1.0	0.983735	0.129330	2.7	2.758153	
5	2.5	32.0	1.0	0.996618	0.061080	2.7	2.736274	
6	2.5	64.0	1.0	0.999226	0.030118	2.7	2.719874	
7	2.5	128.0	1.0	0.999814	0.015015	2.7	2.710382	
8	2.5	256.0	1.0	0.999955	0.007505	2.7	2.705305	
9	2.5	1.0	2.0	0.684071	1.233649	2.7	2.154363	
10	2.5	2.0	2.0	0.144306	1.652012	2.7	-0.881944	
11	2.5	4.0	2.0	0.858119	0.496573	2.7	1.926304	
12	2.5	8.0	2.0	0.994675	0.110136	2.7	2.611660	
13	2.5	16.0	2.0	0.999706	0.026798	2.7	2.681008	
14	2.5	32.0	2.0	0.999982	0.006655	2.7	2.695413	
15	2.5	64.0	2.0	0.999999	0.001661	2.7	2.698863	
16	2.5	128.0	2.0	1.000000	0.000415	2.7	2.699716	
17	2.5	256.0	2.0	1.000000	0.000104	2.7	2.699929	

Out[4]:
```
time                  2.500000
steps               256.000000
order                 2.000000
fidelity              1.000000
state_error           0.000104
exact_gap_proxy       0.803355
trotter_gap_proxy     0.803360
Name: 17, dtype: float64
```

In [5]:
```
show_figure('bcs_trotter_error.png')
```

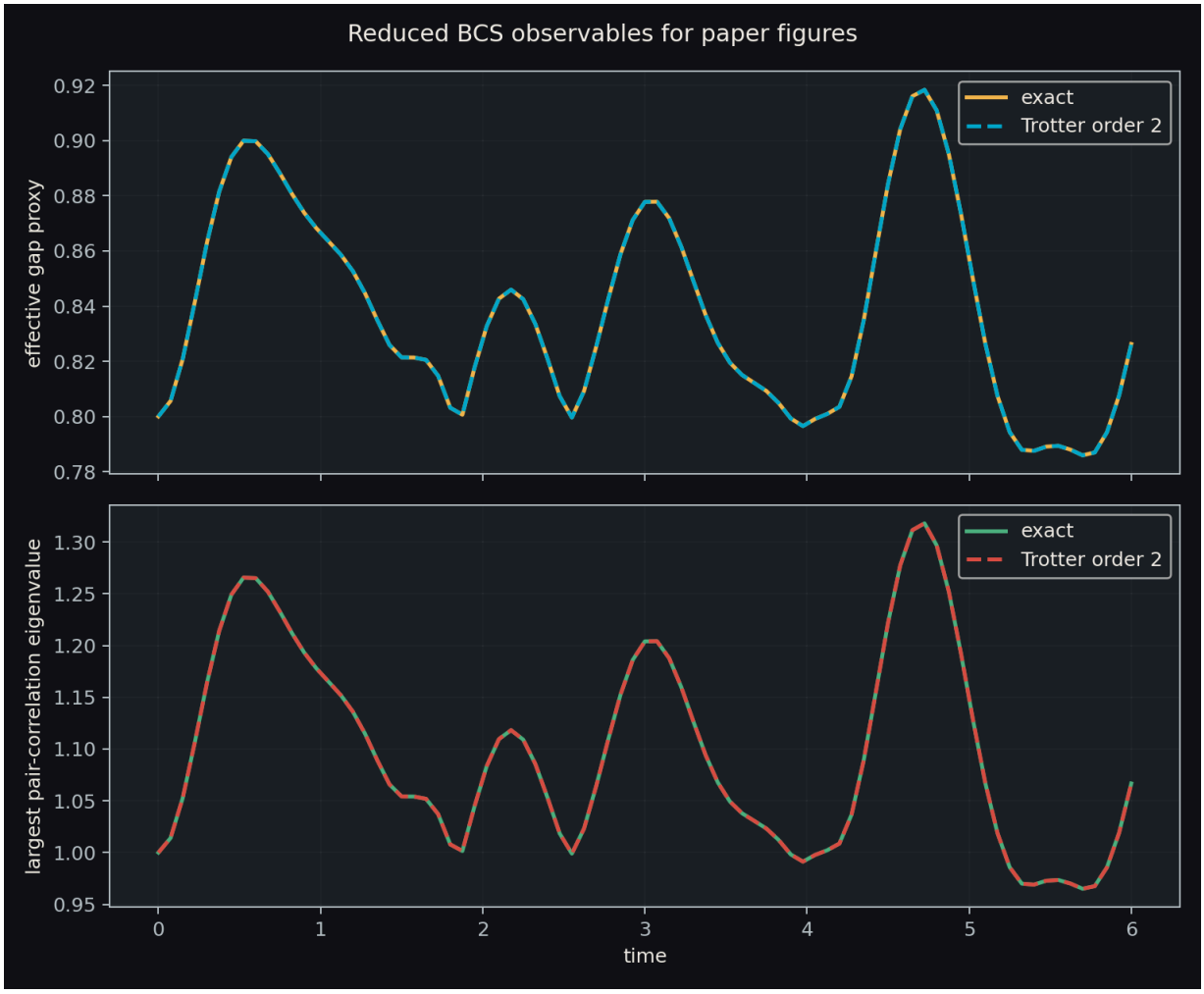


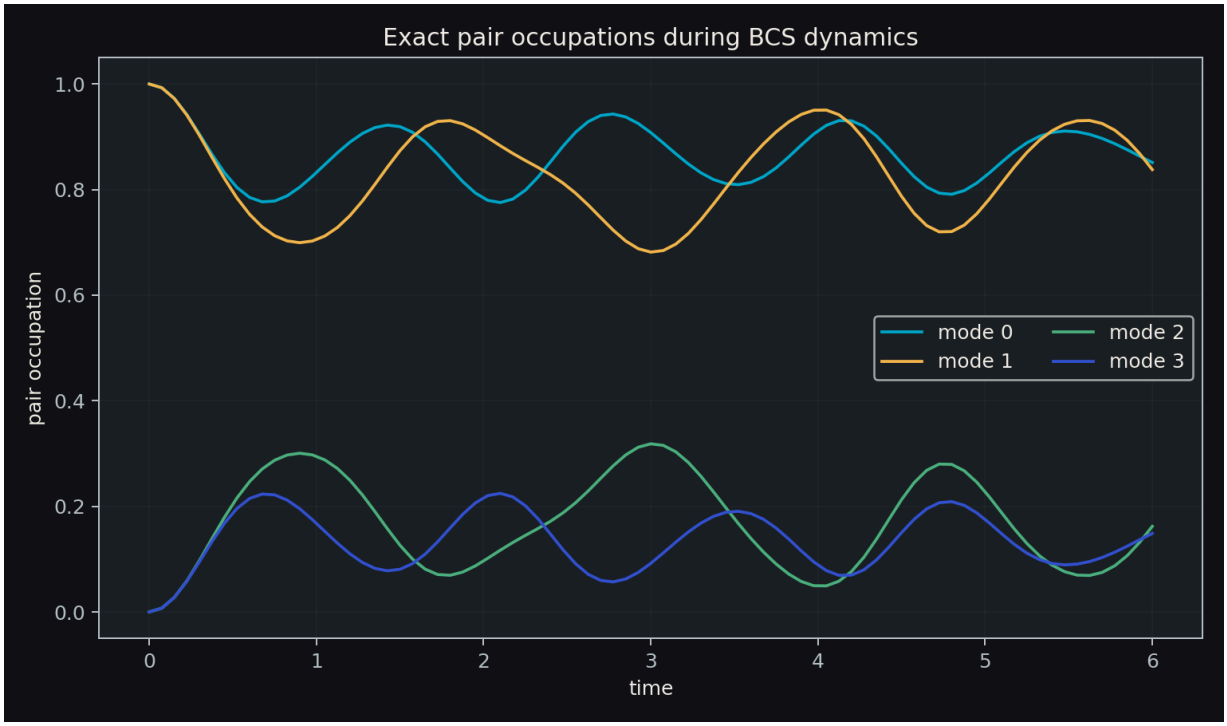
Observáveis físicos para o paper

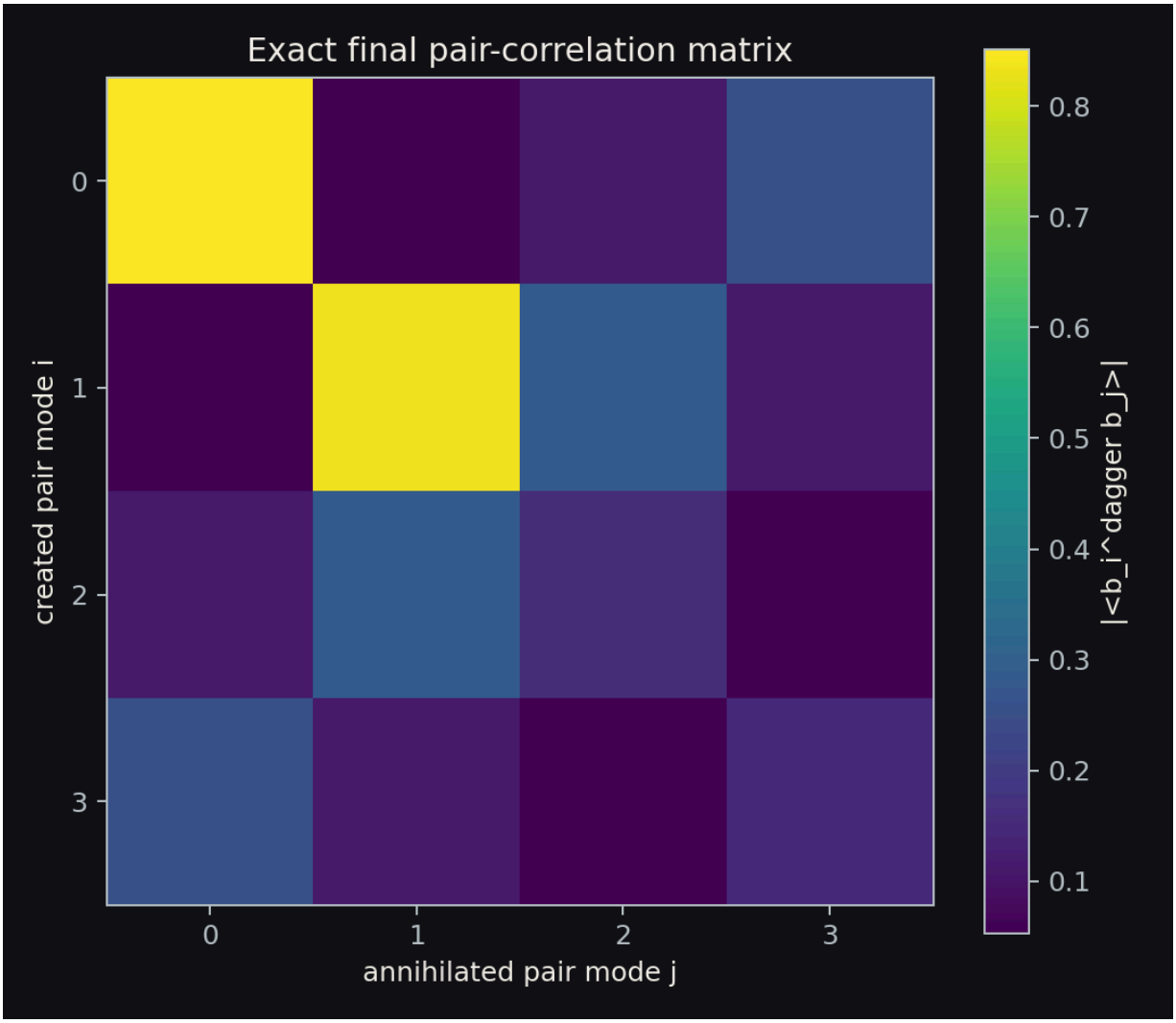
In [6]:
```
obs = read_table('bcs_paper_observables.csv')
final_obs = obs.iloc[-1]
final_obs[['time', 'fidelity', 'state_error', 'exact_pair_number', 'trotter_
```

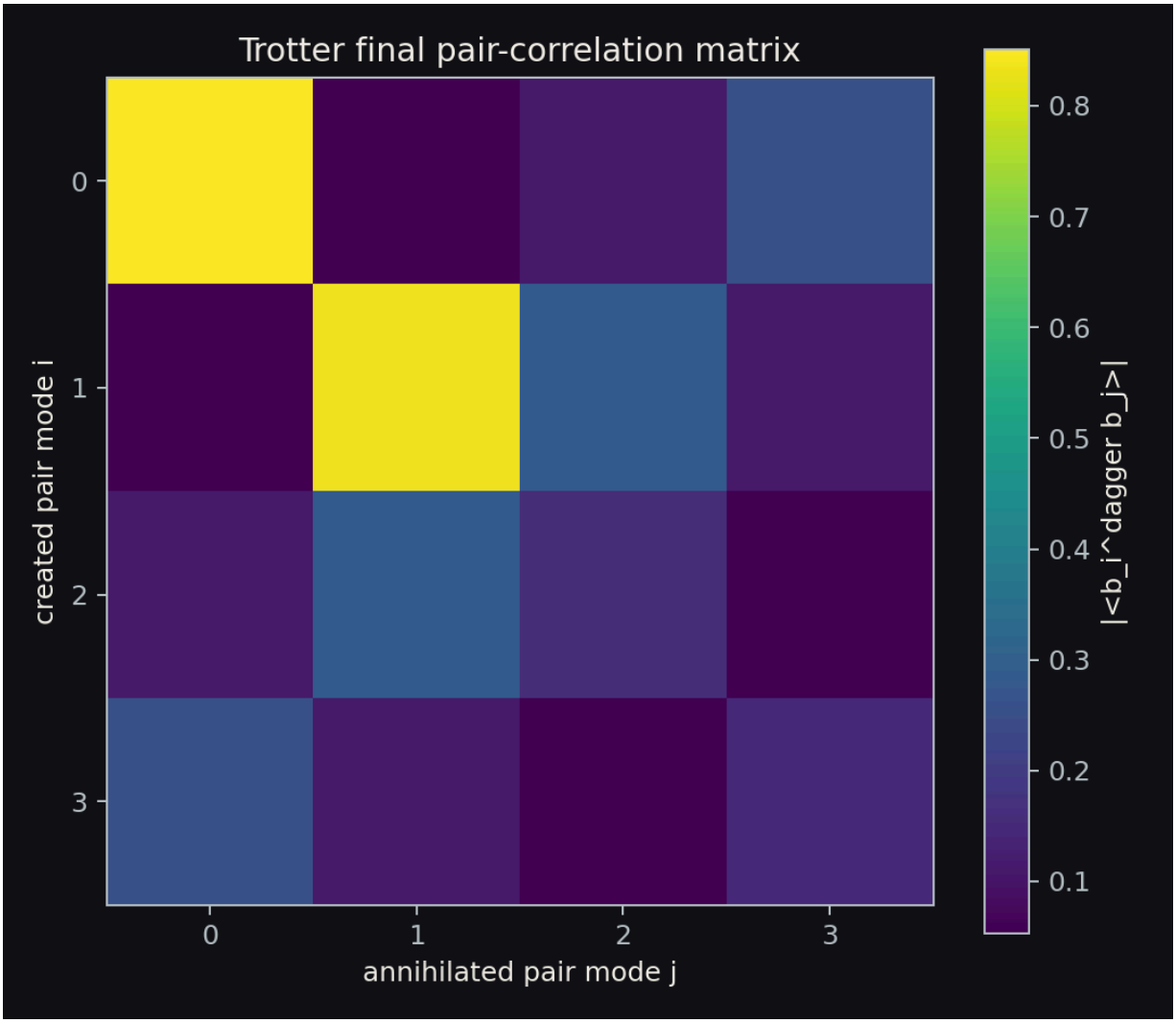
Out[6]:
```
time                 6.000000
fidelity             1.000000
state_error          0.001410
exact_pair_number    2.000000
trotter_pair_number  2.000000
exact_gap_proxy      0.826271
trotter_gap_proxy    0.826182
Name: 80, dtype: float64
```

In [7]:
```
for fig in [
    'bcs_gap_condensate_time.png',
    'bcs_pair_occupations_time.png',
    'bcs_pair_correlation_exact_final.png',
    'bcs_pair_correlation_trotter_final.png',
]:
    show_figure(fig, width=760)
```


Reduced BCS observables for paper figures
effective gap proxy
exact
Trotter order 2
largest pair-correlation eigenvalue
exact
Trotter order 2
time


Exact pair occupations during BCS dynamics
pair occupation
mode 0
mode 1
mode 2
mode 3
time


Exact final pair-correlation matrix
created pair mode i
annihilated pair mode j
|<b_i^dagger b_j>|
0
1
2
3
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1



Green normal, Gor'kov anômalo e matriz Nambu

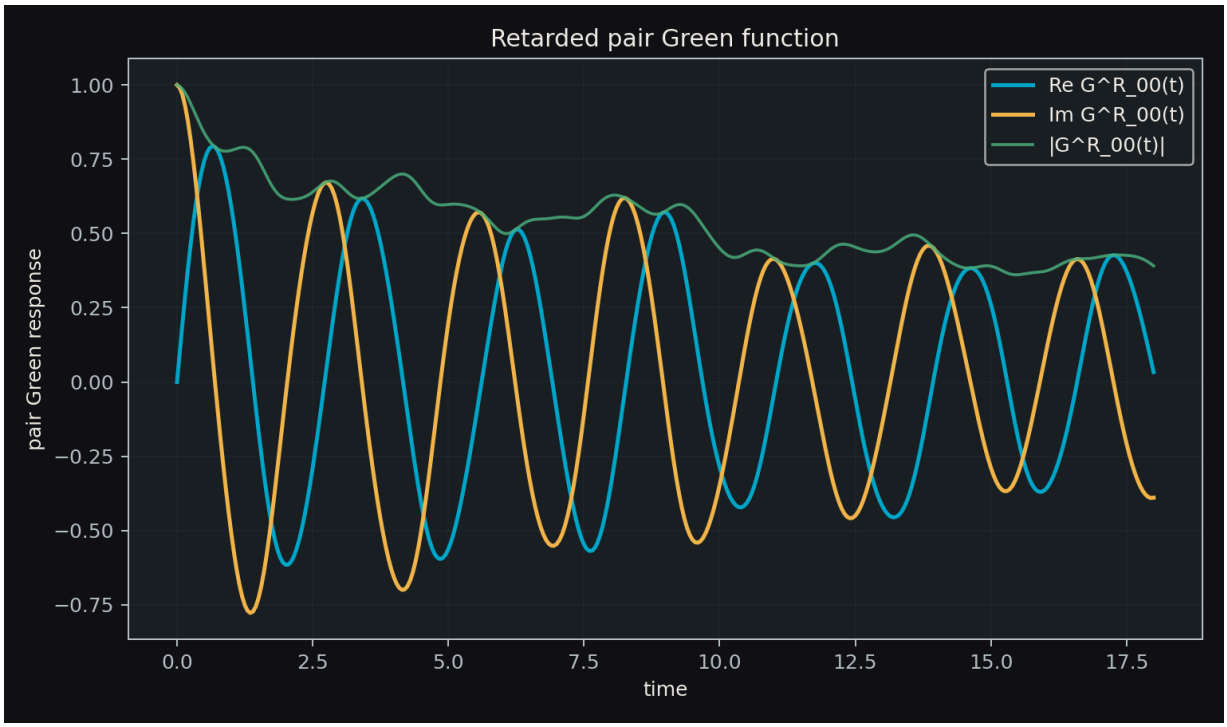
In [8]:
```
green_spec = read_table('bcs_pair_green_spectral_density.csv')
gorkov_spec = read_table('bcs_gorkov_anomalous_spectrum.csv')
nambu_spec = read_table('bcs_gorkov_nambu_spectral_trace.csv')

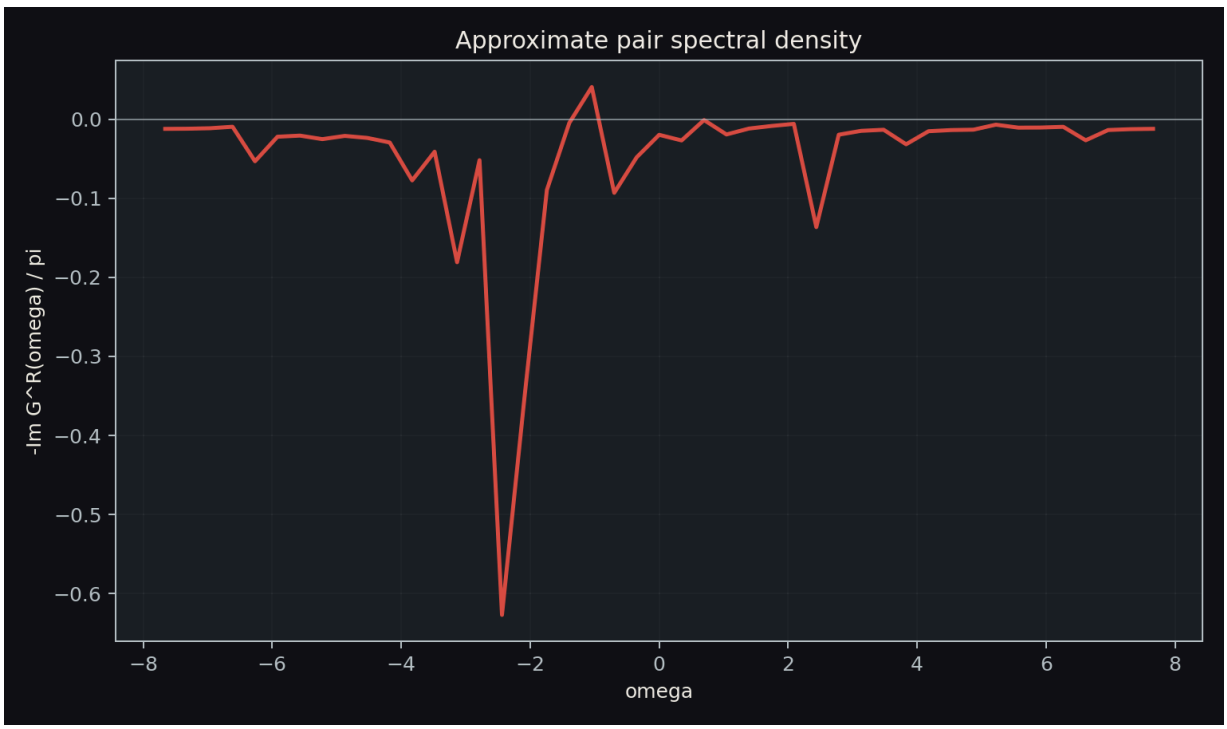
summary_spectral = pd.DataFrame([
    {'channel': 'normal G', **green_spec.sort_values('spectral_weight', asce
    {'channel': 'anomalous F', **gorkov_spec.sort_values('spectral_weight',
    {'channel': 'Nambu trace', **nambu_spec.sort_values('spectral_weight', a
])
summary_spectral
```

Out[8]:

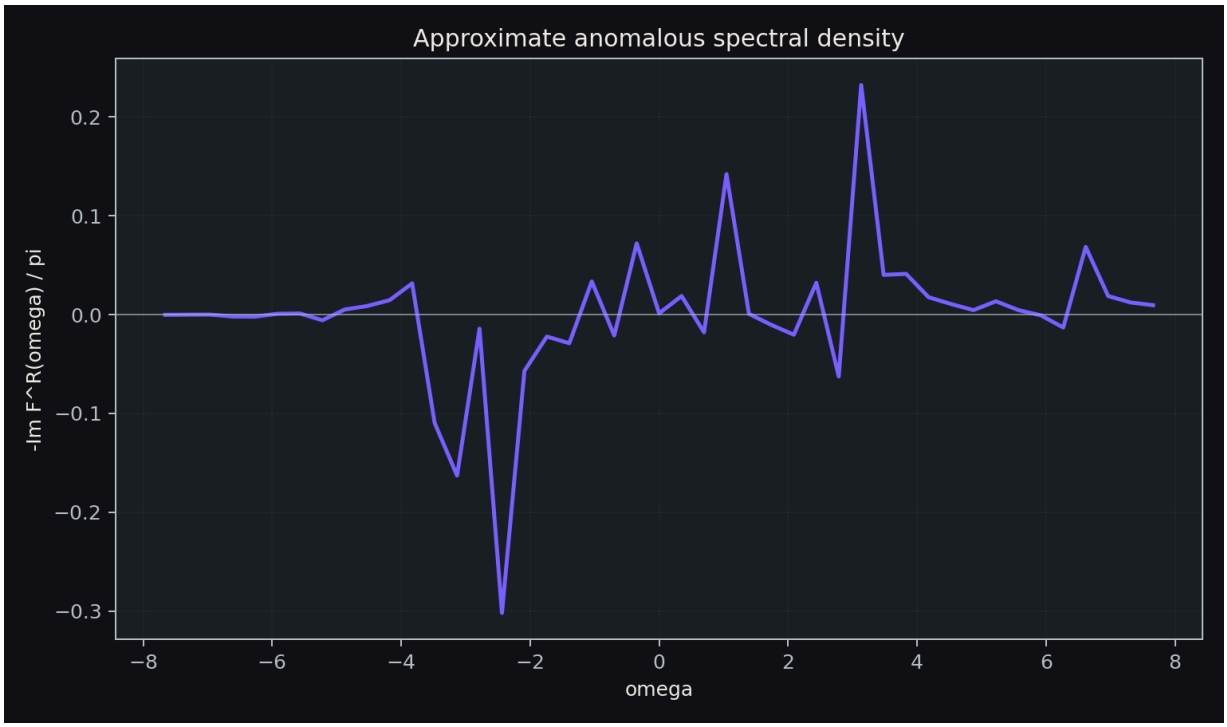
	channel	omega	spectral_weight
0	normal G	-1.044297	0.040784
1	anomalous F	3.132890	0.232452
2	Nambu trace	-2.436692	0.673764

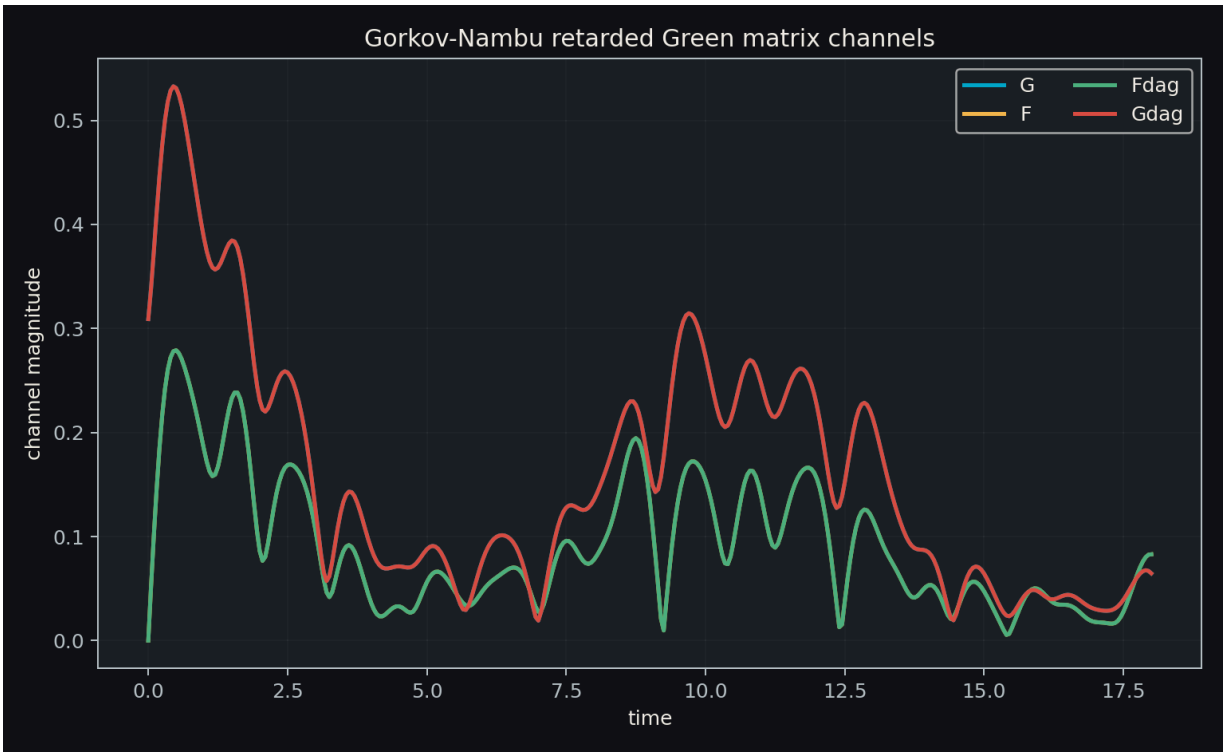
In [9]:
```
for fig in [
    'bcs_pair_green_time.png',
    'bcs_pair_green_spectrum.png',
    'bcs_gorkov_anomalous_time.png',
    'bcs_gorkov_anomalous_spectrum.png',
    'bcs_gorkov_nambu_channels.png',
    'bcs_gorkov_nambu_spectral_trace.png',
]:
    show_figure(fig, width=820)
```

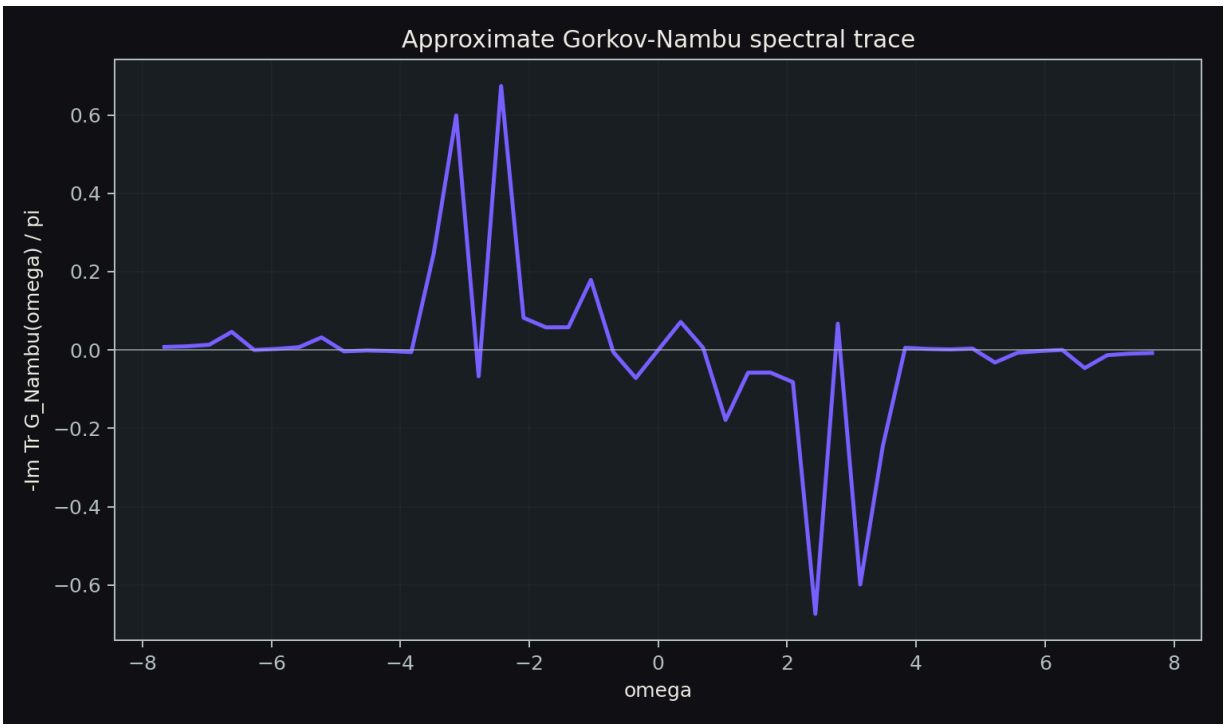





Gorkov anomalous retarded Green function
Re F^R_00(t)
Im F^R_00(t)
|F^R_00(t)|
anomalous pair response
time
0.3
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Approximate anomalous spectral density
-Im F^R(omega) / pi
omega
0.2
0.1
0.0
−0.1
−0.2
−0.3
−8
−6
−4
−2
0
2
4
6
8





Qiskit shots, energia por Pauli e mitigação

In [10]:
```
shots = read_table('bcs_qiskit_shots_noise.csv')
energy = read_table('bcs_qiskit_energy_shots.csv')
mitigation = read_table('bcs_qiskit_error_mitigation.csv')

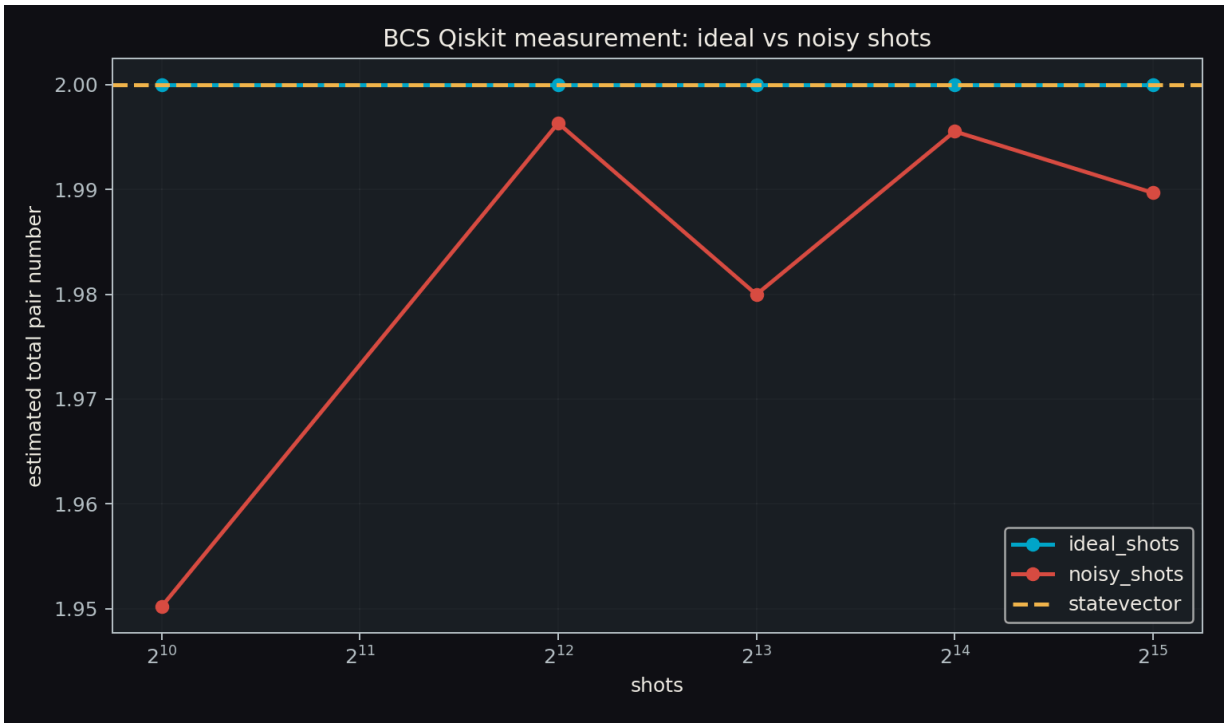
display(shots[['mode', 'shots', 'total_pair_number', 'total_pair_error']])
display(energy[['mode', 'shots', 'exact_energy', 'statevector_energy', 'ener
display(mitigation[['mode', 'noise_scale', 'energy', 'abs_error_vs_statevect
```

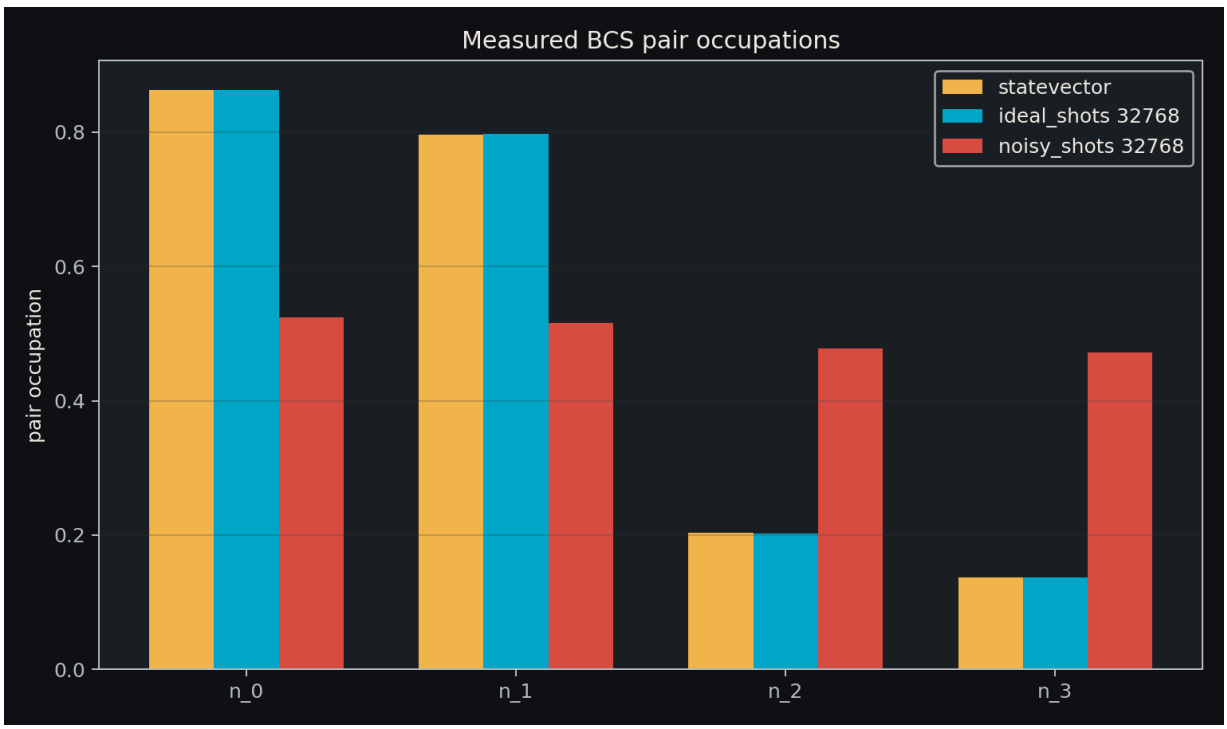
	mode	shots	total_pair_number	total_pair_error
0	ideal_shots	1024.0	2.000000	3.330669e-14
1	noisy_shots	1024.0	1.950195	4.980469e-02
2	ideal_shots	4096.0	2.000000	3.330669e-14
3	noisy_shots	4096.0	1.996338	3.662109e-03
4	ideal_shots	8192.0	2.000000	3.330669e-14
5	noisy_shots	8192.0	1.979980	2.001953e-02
6	ideal_shots	16384.0	2.000000	3.330669e-14
7	noisy_shots	16384.0	1.995544	4.455566e-03
8	ideal_shots	32768.0	2.000000	3.330669e-14
9	noisy_shots	32768.0	1.989685	1.031494e-02

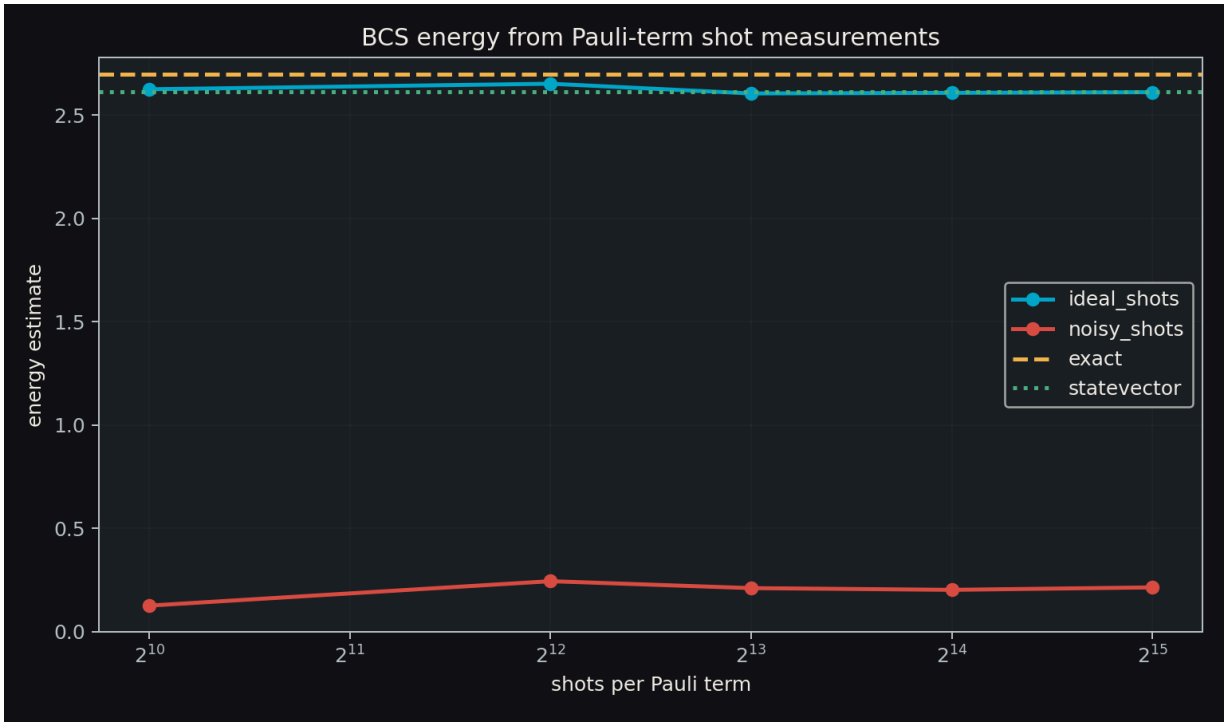
	mode	shots	exact_energy	statevector_energy	energy	abs_error_vs_stateve
0	ideal_shots	1024.0	2.7	2.61166	2.625391	0.013
1	noisy_shots	1024.0	2.7	2.61166	0.125879	2.485
2	ideal_shots	4096.0	2.7	2.61166	2.652319	0.040
3	noisy_shots	4096.0	2.7	2.61166	0.244165	2.367
4	ideal_shots	8192.0	2.7	2.61166	2.605103	0.006
5	noisy_shots	8192.0	2.7	2.61166	0.210583	2.401
6	ideal_shots	16384.0	2.7	2.61166	2.607983	0.003
7	noisy_shots	16384.0	2.7	2.61166	0.202246	2.409
8	ideal_shots	32768.0	2.7	2.61166	2.611313	0.000
9	noisy_shots	32768.0	2.7	2.61166	0.214069	2.397

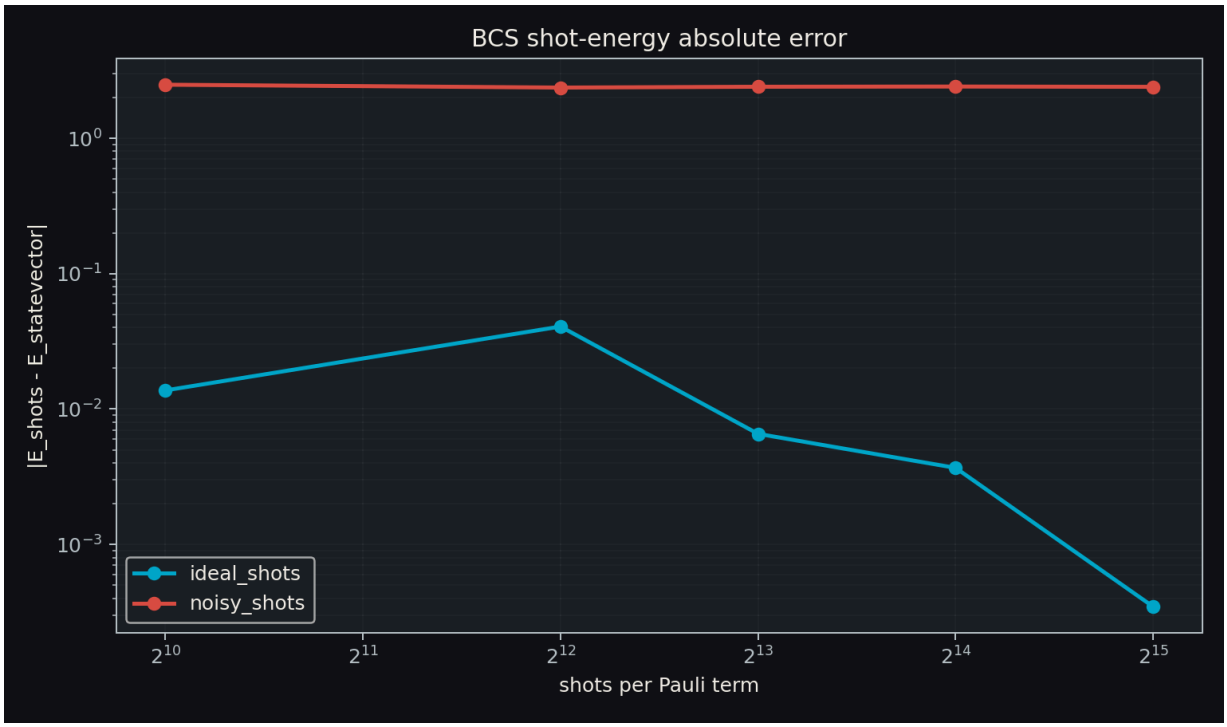
	mode	noise_scale	energy	abs_error_vs_statevector	noise_slope	fit_rmse
0	scaled_noise	0.25	2.219580	0.392080	NaN	NaN
1	scaled_noise	0.50	1.915979	0.695681	NaN	NaN
2	scaled_noise	0.75	1.632208	0.979452	NaN	NaN
3	scaled_noise	1.00	1.404230	1.207430	NaN	NaN
4	linear_zne	0.00	2.475455	0.136205	-1.091929	0.019328

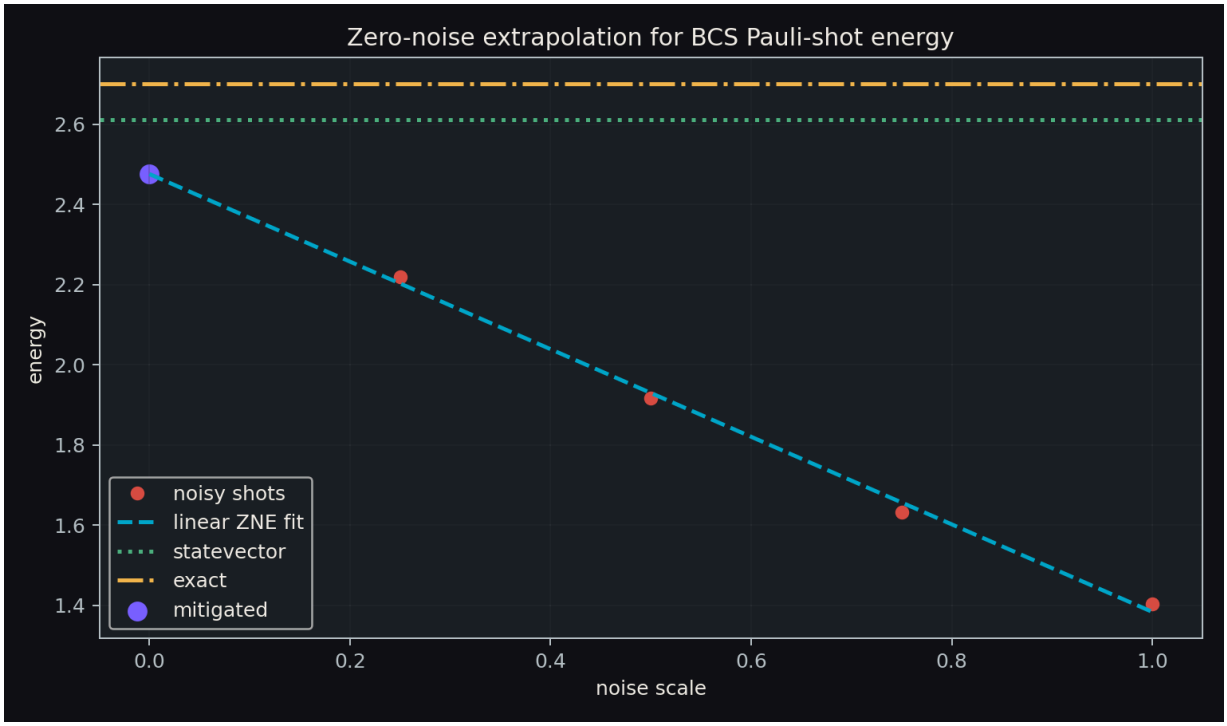
In [11]:
```
for fig in [
    'bcs_qiskit_shots_noise_total_pairs.png',
    'bcs_qiskit_shots_noise_occupations.png',
    'bcs_qiskit_energy_shots_values.png',
    'bcs_qiskit_energy_shots_error.png',
    'bcs_qiskit_error_mitigation_zne.png',
    'bcs_qiskit_error_mitigation_errors.png',
]:
    show_figure(fig, width=820)
```

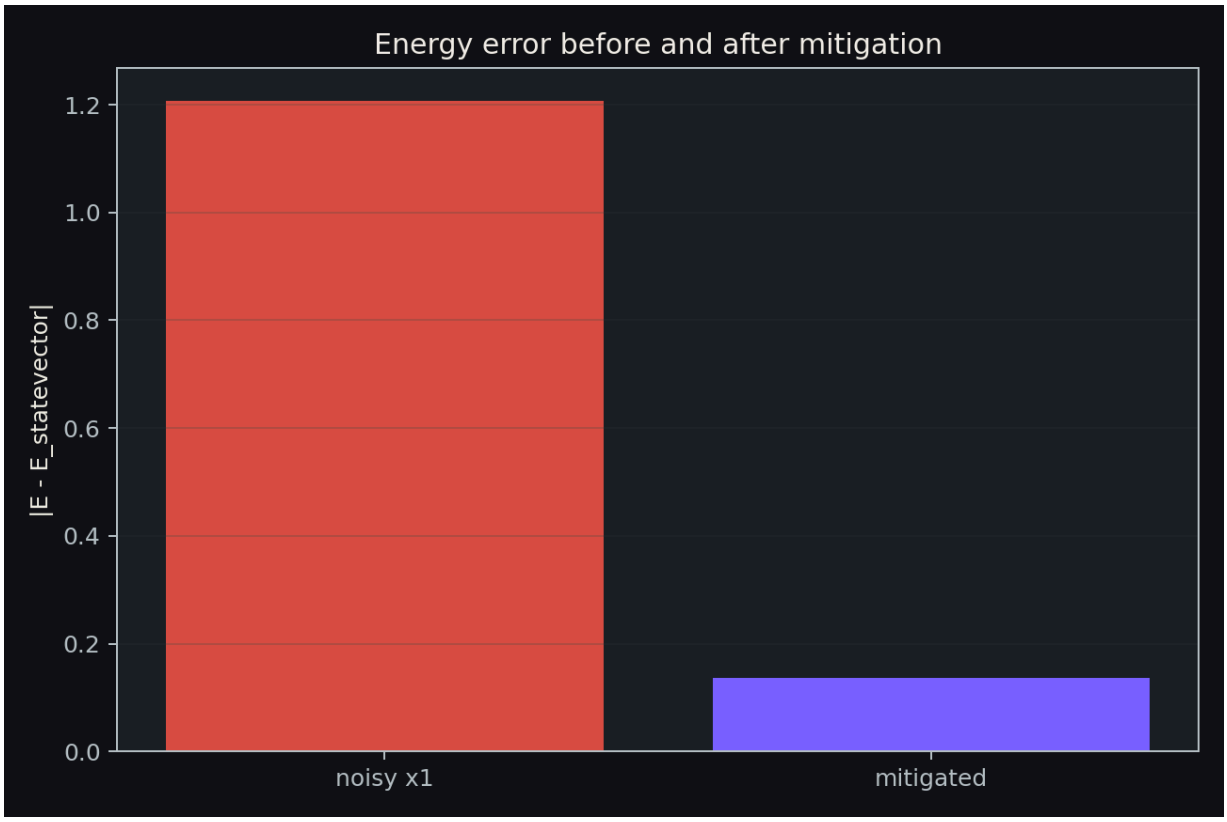





BCS energy from Pauli-term shot measurements
energy estimate
shots per Pauli term
ideal_shots
noisy_shots
exact
statevector
0.0
0.5
1.0
1.5
2.0
2.5
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}


BCS shot-energy absolute error
|E_shots - E_statevector|
shots per Pauli term
ideal_shots
noisy_shots
10^{0}
10^{-1}
10^{-2}
10^{-3}
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}





Resumo numérico final

In [12]:
```
energy_32768 = energy[(energy['mode'] == 'ideal_shots') & (energy['shots'] =
noisy_scale_1 = mitigation[(mitigation['mode'] == 'scaled_noise') & (mitigat
zne = mitigation[mitigation['mode'] == 'linear_zne'].iloc[0]

paper_summary = pd.DataFrame([
    {'metric': 'final observable fidelity', 'value': final_obs['fidelity']},
```

```
    {'metric': 'final state error', 'value': final_obs['state_error']},
    {'metric': 'normal Green peak omega', 'value': summary_spectral.loc[0, '
    {'metric': 'anomalous Green peak omega', 'value': summary_spectral.loc[1
    {'metric': 'Nambu trace peak omega', 'value': summary_spectral.loc[2, 'c
    {'metric': '32768 ideal shot energy error vs statevector', 'value': ener
    {'metric': 'noisy energy error before mitigation', 'value': noisy_scale_
    {'metric': 'energy error after ZNE mitigation', 'value': zne['abs_error_
])
paper_summary
```

Out[12]:

	metric	value
0	final observable fidelity	1.000000
1	final state error	0.001410
2	normal Green peak omega	-1.044297
3	anomalous Green peak omega	3.132890
4	Nambu trace peak omega	-2.436692
5	32768 ideal shot energy error vs statevector	0.000347
6	noisy energy error before mitigation	1.207430
7	energy error after ZNE mitigation	0.136205

Regenerar resultados

Execute no terminal, a partir da raiz do repositório:

```
source .venv/bin/activate
python experiments/run_all.py
pandoc papers/bcs_quantum_simulation.md -s -o
papers/bcs_quantum_simulation.html
google-chrome --headless --disable-gpu --no-sandbox \
  --print-to-pdf="$PWD/papers/bcs_quantum_simulation.pdf" \
  "file://$PWD/papers/bcs_quantum_simulation.html"
```

Para validar o núcleo:

```
PYTHONPATH=src .venv/bin/python -m pytest -q tests
```